

STRABLE: Benchmarking Tabular Machine Learning with Strings

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What is tabular learning?

Customer	Age	Plan	City	Customer comment	Churn
Marie Dupont	34	Premium	Lyon	"billing issue again, second time this month"	no
J. Smith	61	Basic	Paris	"great support, solved in 5 min"	no
Ana König	NA	Premium	Saclay	"still waiting for a callback..."	?

• textual • categorical • numerical • target

- ▶ **Tabular data:** a 2-dimensional arrangement of information; rows are samples, columns are attributes, which have their own type and independent meaning
- ▶ **Tabular learning:** supervised, row-wise prediction of a target, either classification or regression

$$\text{Learn } f : \underbrace{(x^{(1)}, \dots, x^{(m)})}_{\text{heterogeneous features}} \mapsto y \quad \text{from labeled rows}$$

STRABLE fills a gap: evaluating strings "in the wild"

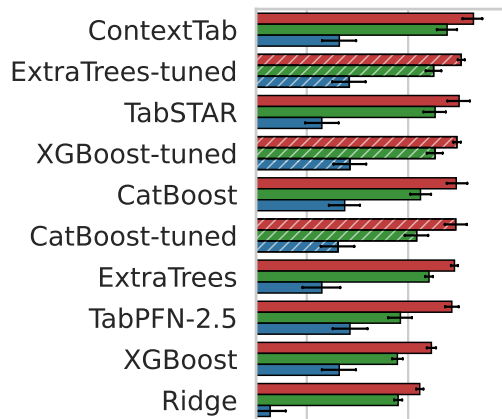
Real-world tables contain strings — names, codes, descriptions — yet existing benchmarks either remove, pre-encode, or ignore them.

- ▶ Focus mainly on **numerical** (OpenML-CC18, PMLB, PMLBmini)
- ▶ **Remove** missing values and high-cardinality categorical features (Grinsztajn et al. 2022, TabReD)
- ▶ **Pre-encode** strings to numbers (AMLB, TabRepo, CARTE)
- ▶ Focus only on **long free-text** (TextTabBench: 13 datasets)
- ▶ Rely on a **single static LLM** (BeyondArena) or demand computationally expensive **task-specific fine-tuning** (MulTaBench)

⇒ **STRABLE**: a corpus of 108 tables with strings, across 8 application fields, with minimal preprocessing.

Strings are pivotal for tabular learning

■ Num+Str ■ Str-only ■ Num-only ▨ Tuned



Avg Score (R^2 & AUC) with 95% CI

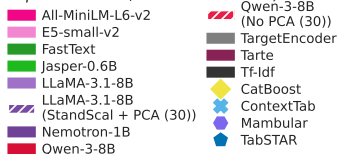
- ▶ **180+ pipelines** evaluated: 9 encoders (6 LLMs, Tf-Idf, TargetEncoder, Tarte), 7 modular learners, 4 end-to-end architectures
- ▶ Same pipelines on **Num-only**, **Str-only**, and **Num+Str** tables
- ▶ Strings carry **independent predictive signal** — raising performance across all learners

+50% increase in average performance from string features vs. numeric-only baselines.

Encoders explain much of the runtime; Modular Pipelines outperform end-to-end architectures

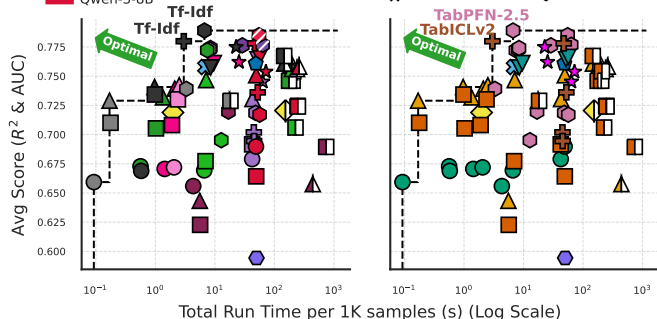
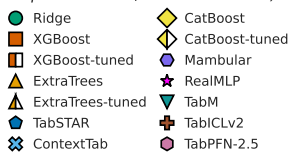
Encoder

(shape = learner, color = encoder)



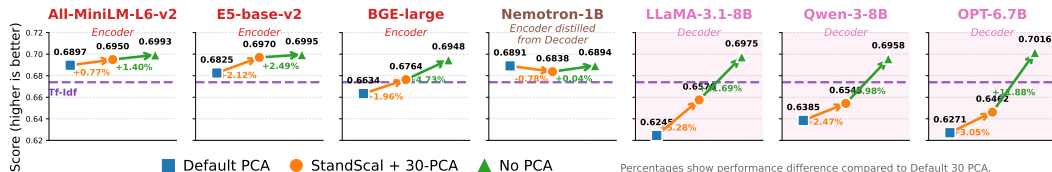
Learner

(shape = learner, color = learner)



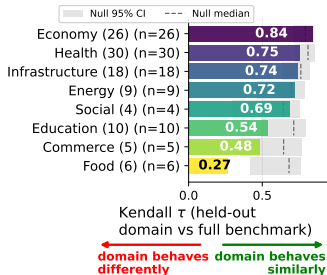
- ▶ Each **cell** is encoded independently
- ▶ LM embeddings extracted via **mean pooling** over the last layer
- ▶ Per text feature, LM embeddings are reduced with a **30-PCA** before passing to the learner

Different LLMs need different post-processing before learning

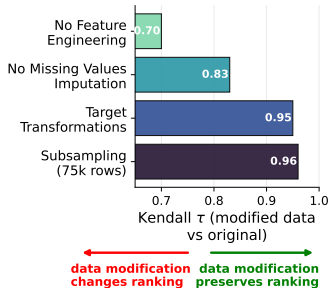


- Encoders tolerate PCA, Decoders don't

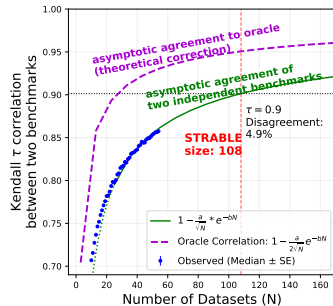
Generalizability and Robustness



- STRABLE is stable across application fields

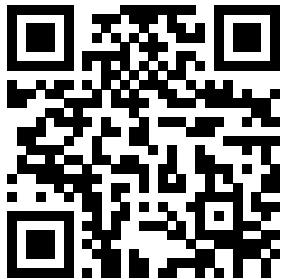


- Benchmark is stable across data preparation choices



- STRABLE has sufficient datasets to approach oracle rankings

Use STRABLE in your research!



soda-inria.github.io/strable

Already evaluated on STRABLE in industry research:

 **PRIOR** TabPFN-3.5



CASE - context-aware semantic embeddings for tabular learners

Public Leaderboard **coming soon** — stay tuned!